

Further Scales and Two-Dimensional Flow

From a probability model to a Navier–Stokes research question

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Question. A finite experiment ends at some length or time scale R . When does a relevant process exist beyond R ? There are two distinct answers. A point-process model can make further scales probable by assumption. A fluid equation can provide a mechanism that transfers activity toward larger scales. Neither answer lets a finite record prove that nature continues forever.

1. A probability law for further scales

Index relevant processes by a characteristic scale $L \geq L_{\min}$. Equal multiplicative steps in L are equal intervals in $\log L$. Assume that relevant scales form a Poisson point process with intensity

$$d\mu(L) = \eta(L) \frac{dL}{L}, \quad \eta(L) \geq 0.$$

This is a population assumption: “relevant” means that a counted process is dynamically coupled to the phenomenon, not merely present elsewhere. If $V(L) = v_d L^d$, then $dV/V = d \, dL/L$; that dimensional factor is absorbed into η , since finite counts identify only the product.

The number $N(R, S)$ of processes with scales in $(R, S]$ is Poisson with mean

$$\mu(R, S) = \int_R^S \eta(L) \frac{dL}{L}.$$

Hence its zero-count probability gives

$$\mathbb{P}\{N(R, S) \geq 1\} = 1 - \exp\left[-\int_R^S \eta(L) \frac{dL}{L}\right]. \quad (1)$$

Letting $S \rightarrow \infty$ yields the sharp criterion

$$\boxed{\text{unbounded process scales almost surely}} \iff \boxed{\int_R^\infty \eta(L) \frac{dL}{L} = \infty}. \quad (2)$$

Coin-flip intuition. If $\eta(L) = \eta_0 > 0$, each doubling band $(2^j R, 2^{j+1} R]$ has the same hit probability

$$p = 1 - \exp(-\eta_0 \log 2) = 1 - 2^{-\eta_0}.$$

The bands are independent. The probability of missing the first m bands is $2^{-m\eta_0} \rightarrow 0$, just as repeated fair coin flips eventually produce heads. In fact, infinitely many bands contain a process almost surely. This is the scale-invariant Poisson process studied by Arratia [1].

The opposite conclusion is equally possible. If $\eta(L) = \eta_0 (L_{\min}/L)^a$ with $a > 0$, then

$$\int_R^\infty \eta(L) \frac{dL}{L} = \frac{\eta_0}{a} \left(\frac{L_{\min}}{R}\right)^a < \infty.$$

There is consequently a positive probability

$$\exp\left[-\frac{\eta_0}{a} \left(\frac{L_{\min}}{R}\right)^a\right]$$

that no relevant process exists beyond R .

Why finite data cannot decide. Choose two intensity functions that agree for every observed $L \leq S_{\text{obs}}$, but give one a constant tail and the other a power-decaying tail. They induce exactly the same observations inside the measured range, while (2) gives opposite answers outside it. The continuation claim therefore comes from the tail model, not from the finite record.

2. What Navier–Stokes can supply

Two-dimensional turbulence supplies a physical route to larger scales. Consider the stochastically forced, damped Navier–Stokes equations on a periodic box $\mathbb{T}_\lambda^2$:

$$du + [(u \cdot \nabla)u + \nabla p]dt = [\nu \Delta u - \alpha(-\Delta)^{-2\gamma}u]dt + dW^\lambda, \quad \nabla \cdot u = 0. \quad (3)$$

The forcing acts near a fixed injection scale and supplies mean energy $\varepsilon > 0$ per unit time and area. Bedrossian, Coti Zelati, Punshon-Smith, and Weber proved rigorous inverse-flux laws for stationary solutions in the large-box limit [2]. Their hypotheses include regular forcing, $\lambda \rightarrow \infty$, and weak anomalous dissipation as $\nu, \alpha \rightarrow 0$:

$$\nu \mathbb{E} \|\omega\|_2^2 \rightarrow 0, \quad \alpha \mathbb{E} \|(-\Delta)^{-\gamma} \omega\|_2^2 \rightarrow 0, \quad \omega = \text{curl } u. \quad (4)$$

Here the exponent -2γ in the velocity equation becomes $-\gamma$ in the quadratic dissipation norm; this is the convention of Ref. [2].

For the velocity increment $\delta_{\ell n} u(x) = u(x + \ell n) - u(x)$, define

$$\mathcal{F}_\ell(u) = \int_{S^1} \int_{\mathbb{T}_\lambda^2} |\delta_{\ell n} u|^2 (\delta_{\ell n} u \cdot n) dx dn.$$

There is an outer inertial scale $\ell_\alpha \rightarrow \infty$ such that

$$\mathbb{E} \mathcal{F}_\ell(u) \sim 2\varepsilon \ell, \quad 1 \ll \ell \ll \ell_\alpha. \quad (5)$$

The positive sign is the inverse cascade: energy crosses toward larger lengths. Thus, for every finite R , sufficiently large boxes and sufficiently weak drag admit an active inverse-flux scale $\ell \geq R$.

For each fixed box, viscosity, and positive drag, suppose additionally that the chosen invariant measure is ergodic. Birkhoff’s theorem then gives

$$\frac{1}{T} \int_0^T \mathcal{F}_\ell(u(t)) dt \longrightarrow \mathbb{E} \mathcal{F}_\ell(u) > 0. \quad (6)$$

Thus positive inverse transfer recurs at each admitted scale almost surely. This stationary result concerns arbitrarily large scales only across a large-box, vanishing-drag family; a fixed box still has a largest scale.

3. The stronger theorem still wanted

The desired single-flow statement would construct a homogeneous statistical solution on $\mathbb{R}^2$, with finite energy per unit area (not a finite-total- energy $L^2(\mathbb{R}^2)$ solution). Let $E(k, t)$ be its isotropic energy spectrum and, for fixed $\theta \in (0, 1)$, define the infrared energy-quantile length

$$k_\theta(t) = \inf \left\{ K > 0 : \int_0^K E(k, t) dk \geq \theta \int_0^\infty E(k, t) dk \right\}, \quad L(t) = k_\theta(t)^{-1}.$$

The target is then the well-posed growth statement

$$\mathbb{P}\{L(t) \rightarrow \infty\} = 1. \quad (7)$$

The familiar rate follows if one additionally assumes the Kraichnan inverse-cascade spectrum [3]

$$E(k) = C_I \varepsilon^{2/3} k^{-5/3}, \quad L(t)^{-1} \leq k \leq L_f^{-1}.$$

Indeed, equating accumulated energy εt with the spectral integral gives

$$\varepsilon t = \frac{3C_I}{2} \varepsilon^{2/3} (L(t)^{2/3} - L_f^{2/3}),$$

so

$$L(t) = \left[L_f^{2/3} + \frac{2}{3C_I} \varepsilon^{1/3} t \right]^{3/2} \sim \left(\frac{2}{3C_I} \right)^{3/2} \varepsilon^{1/2} t^{3/2}. \quad (8)$$

Equation (8) is still a closure calculation, not a theorem derived from (3). Here L is an infrared-front proxy (different fixed θ change constants, not the predicted exponent). This evolving regime is nonstationary, so the fixed-parameter Birkhoff argument in (6) does not imply (7) or (8). The missing step is to prove an expanding infrared front or the $k^{-5/3}$ spectrum directly from Navier–Stokes on an unbounded domain; homogeneous forcing makes this harder because total energy is infinite. Large-scale drag, finite forcing duration, or a finite container can stop the growth. The research target is to establish (7), with explicit hypotheses and cutoffs, without assuming the spectrum that already implies it.

References

- [1] R. Arratia, “On the Central Role of the Scale Invariant Poisson Processes on $(0, \infty)$,” DIMACS Series **41** (1998), [arXiv:1611.05572](#).
- [2] J. Bedrossian, M. Coti Zelati, S. Punshon-Smith, and F. Weber, “Sufficient Conditions for Dual Cascade Flux Laws in the Stochastic 2d Navier–Stokes Equations,” *Arch. Ration. Mech. Anal.* **237** (2020), [103–145](#).
- [3] R. H. Kraichnan, “Inertial Ranges in Two-Dimensional Turbulence,” *Phys. Fluids* **10** (1967), [1417–1423](#).